

MUHAMMAD FAHEEM ARSHAD

AI Solution Architect & Tech Lead | Node.js · JavaScript · Full-Stack · AI/LLM Engineering

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PROFESSIONAL SUMMARY

Results-driven Solution Architect and Tech Lead with 10+ years of experience designing and delivering scalable Node.js and JavaScript applications across backend and frontend domains. Deep expertise in NestJS, Express.js, TypeScript, React.js, Next.js, Angular, and REST/GraphQL API development. Proven track record building high-performance B2B AI-powered SaaS platforms, microservices architectures, and enterprise web applications. Skilled in cloud infrastructure (AWS, Azure), SQL/NoSQL databases, queue systems (RabbitMQ/SQS), Elasticsearch, CI/CD pipelines, and Agile/Scrum methodologies. Effective engineering leader who mentors teams, drives code-quality standards, and aligns technical strategy with business objectives.

CORE COMPETENCIES

Backend Development Node.js · NestJS · Express.js · REST APIs · GraphQL · Microservices · WebSockets	Frontend Development TypeScript · React.js · Next.js · Angular · AngularJS · HTML5 · CSS3 · Tailwind
Databases & Search MySQL · PostgreSQL · MongoDB · Elasticsearch · Redis · Database Design & Optimization	Cloud & DevOps AWS (EC2, S3, Lambda, SQS, SNS) · Azure · Docker · CI/CD · GitHub Actions · Datadog
Architecture & System Design Solution Architecture · Microservices · DDD · Event-Driven · Queue Systems (RabbitMQ / SQS) · Browser Automation (Playwright · Puppeteer)	AI & LLM Engineering Claude (Anthropic) · Gemini · Multi-LLM Orchestration · Prompt Engineering · AI Agent Design · Reviewer Agents · Data Enrichment Pipelines · Browser AI Automation
AI-Assisted Development Cursor IDE · GitHub Copilot · AI-Augmented Debugging · LLM-Powered Code Review · Rapid Prototyping	Leadership & Process Tech Lead · Team Management · Agile · Scrum · Code Reviews · Mentoring · Technical Roadmapping · Stakeholder Communication

PROFESSIONAL EXPERIENCE

Team Lead – AI Engineering Refocus AI — Remote	<i>Mar 2025 – Present</i>
- Led an AI engineering team delivering browser automation solutions targeting the USA insurance market — scraping, navigating, and extracting structured data from carrier portals at scale using Node.js, Playwright, and Puppeteer. - Architected a multi-source AI pipeline dynamically routing tasks across Claude (Anthropic), Gemini, and other LLMs based on cost, latency, and accuracy demand — reducing inference cost while maintaining output quality. - Designed and implemented data enrichment workflows integrating paid third-party APIs (address validation, carrier lookups, demographic enrichment) to enhance raw records before downstream processing. - Built reviewer agent layers that autonomously validate and score AI-generated outputs, flagging low-confidence results for human review and improving end-to-end data accuracy. - Leveraged Cursor AI-assisted IDE for accelerated coding, refactoring, and bug-fixing across the Node.js/TypeScript codebase — significantly reducing time-to-fix on production issues. - Created and maintained Datadog dashboards for real-time observability of automation pipelines, agent health, API error rates, and throughput SLAs — enabling proactive incident response. - Designed overall system architecture covering microservices, queue-based task distribution (RabbitMQ/SQS), browser farm orchestration, and multi-agent AI coordination layers.	
Managing Consultant / Tech Lead	<i>Jul 2023 – Mar 2025</i>

Systems Ltd — Lahore

- Led full-stack development of a high-traffic news platform (millions of monthly users) using Node.js, NestJS, Next.js, and MySQL — handled 10k+ concurrent users with horizontal scaling.
- Built multiple CRM products with Angular and RESTful Node.js APIs, delivering features on 2-week Agile sprints.
- Directed a cross-functional engineering team of 8+, running daily standups, sprint planning, and technical roadmap sessions.
- Produced comprehensive API documentation and system-design specs to accelerate onboarding and reduce knowledge silos.
- Analyzed business requirements and translated them into scalable backend microservices and frontend component libraries.

Principal Software Engineer

Jan 2021 – Jul 2023

Confiz Ltd — Lahore

- Spearheaded backend development with Node.js, Express.js, and NestJS following Domain-Driven Design principles.
- Built reusable React.js and Angular component libraries shared across five product lines, accelerating frontend delivery.
- Led knowledge-transfer sessions on TypeScript, async/await patterns, and Node.js performance profiling.
- Conducted in-depth code reviews enforcing SOLID principles and security best practices across backend APIs.
- Introduced queue-based architecture (RabbitMQ) for asynchronous task processing, reducing server load by 30%.

Senior Software Engineer

Feb 2020 – Jan 2021

Strategic Systems International (SSI) — Lahore

- Developed and maintained backend services in Node.js / Express.js integrated with SQL Server and MySQL databases.
- Enforced coding standards and mentored junior JavaScript developers through peer reviews and pair programming.
- Implemented Agile/Scrum rituals (sprint planning, retrospectives) that improved on-time delivery rate by 25%.
- Communicated directly with international clients on API contracts, acceptance criteria, and deployment timelines.

Software Engineer

Oct 2016 – Jan 2020

Dubzil Labs (Zameen.com) — Lahore

- Developed JavaScript / AngularJS frontend features and Node.js backend endpoints for one of Pakistan's largest real-estate portals.
- Participated in MySQL database schema design and query optimization, improving page-load performance by 20%.
- Conducted code refactoring and regression testing during deployments to maintain platform stability.
- Maintained and modernized legacy codebase, progressively migrating AngularJS views to Angular.

Software Engineer

Jan 2016 – Oct 2016

Stellar Technologies — Lahore

- Built JavaScript-based web features and documented software solutions supporting rapid team onboarding.
- Maintained legacy applications and implemented process optimizations, improving development efficiency.

EDUCATION

Bachelor of Science in Computer Science (Honors)

2012 – 2016

Bahauddin Zakariya University — Multan

- Strong foundation in object-oriented programming, data structures, algorithms, databases, and networking.
- Hands-on software development projects covering full-stack web and systems programming.

CERTIFICATIONS & COURSES

Leadership Development Course — LUMS (2024)